

Multiple Choice (10 marks)

1. Complete bilateral cleft palate is due to failure of fusion of the
 - (a) palatine processes.
 - (b) palatine and frontonasal processes.
 - (c) palatine, frontonasal and secondary nasal processes.
 - (d) palatine, frontonasal, secondary nasal processes and mandibular processes.
2. Mapping of human chromosomes:
 - (a) has been restricted to the sex chromosomes because of small family sizes
 - (b) proceeded much more successfully as large numbers of DNA markers became available.
 - (c) has determined that the number of linkage groups is about twice the number of chromosomes
 - (d) has demonstrated that almost all of the DNA is involved in coding for genes
3. The major question concerning aging and personality is:
 - (a) Do older adults still have personalities?
 - (b) Does personality change with age?
 - (c) Does personality decline as we grow older?
 - (d) Does personality increase or decrease with age?
4. Which of the following statements about heritability are true?
 - (a) is a measure of the level of heterozygotes in a population
 - (b) is a measure of the proportion of variation that is due to genetic causes
 - (c) is a measure of proportion of repeated DNA in an organism
 - (d) is a measure of level of gene linkage
5. Fertilization of an ovum by a spermatozoon occurs in which of the following structures?
 - (a) Cervix
 - (b) Fallopian tube
 - (c) Ovary
 - (d) Uterus

True / False (10 marks)

Answer True or False

6. True or False: Congenital adrenal hyperplasia is a condition that can cause female pseudohermaphroditism due to excess androgen production.
7. True or False: Older adults often quit searching for memories because they prioritize more important tasks over memory retrieval.
8. True or False: Oxygen is found in greater quantity in exhaled air than nitrogen, which is a misconception about respiratory gases.
9. True or False: Mutations that cause achondroplasia are classified as gain-of-function mutations, leading to abnormal bone growth.
10. True or False: Older adults generally perform very well when their semantic memory is tested.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of togaviruses in human diseases and critically analyze why Yellow Fever is not classified among the diseases caused by these viruses.
12. Discuss the role of various hormones in the aging process, specifically analyzing why viaseterol has not been recognized as a potential agent against senescence.

End of Paper